

Finding the Whole or the Percent Rate

Rates from a part and a whole, wholes from a part and a rate, and solving the percent equation

Directions: every percent problem holds three numbers — a **part**, a **whole**, and a **rate**. Two are given; find the third.

- **Worked pattern (rate).** 9 of the 30 chairs are taken. $\frac{\text{part}}{\text{whole}} = \frac{9}{30} = 0.3$, and $0.3 \times 100 = 30$, so 30% are taken.
- **Worked pattern (whole).** 9 chairs is 30% of the room. $\frac{30}{100}x = 9$, so $x = 9 \div 0.30 = 30$ chairs.
- Show the division or the equation you used — an answer with no setup earns no credit. Percents may come out as decimals; that is fine.

1. A minibus has 40 seats and 14 of them are taken. What percent of the seats are taken?

Answer: _____

2. A quiz has 24 questions. Dev answered 18 of them correctly. What percent did he answer correctly?

Answer: _____

3. A theatre has already sold 12 tickets, and that is 25% of all the tickets it printed. How many tickets did it print?

Answer: _____

4. Of the 45 seedlings in a tray, 27 have flowered. What percent have flowered?

Answer: _____

5. A hiker has walked 63 kilometres, which is 70% of the whole trail. How many kilometres long is the trail?

Answer: _____

6. In a class survey, 15 students chose soccer, and that is 60% of the class. Write the percent equation $\frac{60}{100}x = 15$ and solve it for the class size x .

Answer: _____

7. A jar holds 56 marbles and 7 of them are green. What percent of the marbles are green?

Answer: _____

8. A cart holds several books; 45 of them are non-fiction, and that is 12.5% of the cart. How many books are on the cart?

Answer: _____

9. A shop sold 91 shirts last month, which was 35% of the shirts it had in stock. Write the percent equation $\frac{35}{100}x = 91$ and solve it for the size x of the stock.

Answer: _____

10. A bird count recorded 60 birds, and 17 of them were finches. What percent of the birds counted were finches? Round to the nearest tenth of a percent.

Answer: _____

11. A deposit of 84 dollars is 20% of the price of a bicycle.

(a) the full price of the bicycle, in dollars: _____

(b) the amount still owed after the deposit, in dollars: _____

12. A survey reported that 45 students walk to school, and that is 30% of the students who were asked.

- (a) Write the percent equation $\frac{30}{100}x = 45$ and solve it for the number of students asked:

- (b) Explain why dividing 45 by 0.30 gives the same number as solving that equation. Write your explanation in the work space.